MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **WEEKLY TEST – 02** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 8: Introduction to Trigonometry** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided where indicated.
2. **Section A** comprises 4 Multiple Choice Questions and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Short Answer Type-I question of 2 marks.
4. **Section C** comprises 3 Short Answer Type-II questions of 3 marks each.
5. **Section D** comprises 1 Applied Case-Based question of 4 marks with internal subparts.

SECTION A: Objective Type Questions (5 × 1 = 5 Marks)

Q1. If $\sin\theta = x$ and $\sec\theta = y$, then $\tan\theta$ is equal to: [1]

(A) xy (B) $\frac{x}{y}$ (C) $\frac{y}{x}$ (D) $\frac{1}{xy}$

Q2. In a right-angled triangle ABC, right-angled at C, the value of $\cos(A+B)$ is: [1]

(A) 0 (B) 1
(C) $\frac{1}{2}$ (D) $\frac{\sqrt{3}}{2}$

Q3. If $\sin A = \frac{1}{2}$, then the value of $\cot A$ is: [1]

(A) $\sqrt{3}$ (B) $\frac{1}{\sqrt{3}}$
(C) $\frac{\sqrt{3}}{2}$ (D) 1

Q4. If $\tan\theta = \frac{a}{b}$, then the value of $\frac{a\sin\theta - b\cos\theta}{a\sin\theta + b\cos\theta}$ is: [1]

(A) $\frac{a^2-b^2}{a^2+b^2}$
(B) $\frac{a^2+b^2}{a^2-b^2}$
(C) $\frac{a}{a^2+b^2}$
(D) $\frac{b}{a^2+b^2}$

Q5. **Assertion (A):** If $\tan A = \frac{4}{3}$, then $\sec A = \frac{5}{3}$.
Reason (R): $\sec^2 A = 1 + \tan^2 A$, so $\sec A = \sqrt{1 + (4/3)^2} = \sqrt{25/9} = \frac{5}{3}$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Short Answer Type-I (1 × 2 = 2 Marks)

Q6. In ΔABC, right-angled at B, $AB = 24$ cm and $BC = 7$ cm. Determine the values of: (i) $\sin A, \cos A$, (ii) $\sin C, \cos C$. [2]

SECTION C: Short Answer Type-II (3 × 3 = 9 Marks)

Q7. Prove that: $\frac{1+\sec A}{\sec A} = \frac{\sin^2 A}{1-\cos A}$. [3]

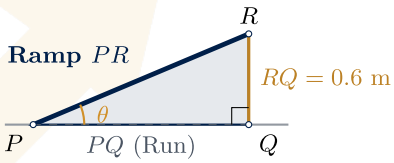
Q8. Prove that: $\sqrt{\frac{1+\sin A}{1-\sin A}} = \sec A + \tan A$. [3]

Q9. In a right ΔABC, $\angle B = 90°$, $BC = 5$ cm and $AC - AB = 1$ cm. Evaluate $\frac{1+\sin C}{1+\cos C}$. [3]

SECTION D: Case-Based / Applied Situation (1 × 4 = 4 Marks)

Q10. ADA Compliant Wheelchair Access Ramp Architecture

According to architectural accessibility standards, an entrance ramp for wheelchairs must have a gentle incline. An architect designs a straight access ramp PR to bridge a vertical doorway threshold rise of $RQ = 0.6$ m above the ground level. The ramp is inclined at an angle α to the horizontal ground such that $\tan\alpha = \frac{1}{12}$.



Based on the above geometric model, answer the following questions:

(i) Determine the horizontal run PQ of the access ramp along the ground. [1]

(ii) Calculate the exact slant length PR of the ramp surface (leave in surd form or round to two decimal places). [1]

(iii) If the ramp slope is redesigned to an angle of $30°$ for a secondary emergency ramp of the same vertical rise 0.6 m, find the new required ramp surface length. [2]

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